

Partition Logics and Linear Orthosets: Representations, Obstructions, and Completions

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Partition logics and linear orthosets both organize mutually exclusive outcomes but encode different data. The former start from a finite carrier and designated partitions; the latter retain only orthogonality and, under global linearity at rank at least four, yield the projective geometry of an anisotropic Hermitian space. We describe the forgetful passage between these structures and distinguish set, projection, ray, and full projective representations. A separating-state criterion characterizes faithful finite partition representations. A rank- n unique-complement obstruction restricts faithful ray representations, with the three-context triangle as its smallest instance. We further separate test saturation, orthogonal closure, and point enlargement as inequivalent completions. This comparison clarifies the roles of finiteness and rank, and why generalized Hermitian geometry does not by itself select a standard real or complex Hilbert space.

I. INTRODUCTION: TWO WAYS OF DESCRIBING EXPERIMENTS

Suppose that an experiment has several possible settings. At each setting the apparatus displays exactly one among a finite list of mutually exclusive outcomes. There are at least two quite different ways to turn this operational description into mathematics.

The first way asks whether the observed distinctions can be modelled by a finite “catalogue of underlying cases.” Let S be that catalogue. Its individual elements need not be operationally accessible: an experiment may reveal only that the actual case belongs to a certain class. In this epistemic sense the elements of S may be called *hidden*, without implying any particular microscopic dynamics or ontological interpretation. Each setting defines an equivalence relation: two elements of S are equivalent when that setting cannot distinguish them by its observable output. The equivalence classes form a partition of S . They are the elementary outcomes available at that setting, and arbitrary unions of classes form a finite Boolean algebra. Several settings give several Boolean algebras, which may overlap. Their union, with the compatible Boolean operations retained within each setting, is a *partition logic*. Generalized urns and finite automata give concrete physical and computational realizations of this construction [1–3].

The general definition allows different settings to have different numbers of outcomes. For the comparison with fixed-dimensional quantum systems developed below, however, we shall concentrate on n -uniform families: every selected partition has exactly n equivalence classes. In ordinary language, every experiment then has the same number n of possible outcomes. This mirrors complete nondegenerate projective measurements, or orthonormal bases, in dimension n . It is a restriction chosen for that comparison, not part of the definition of partition logic

and not a property of quantum measurements in full generality.

In less formal language, a partition logic begins with a finite collection of fully specified cases and then restricts which questions may be asked about them. Non-Boolean structure can arise from the restriction and pasting of questions even though the underlying catalogue remains classical. This is why partition logics can exhibit complementarity without thereby exhibiting Kochen–Specker value indefiniteness.

The second way discards the catalogue, the settings, and even the Boolean operations. It retains only a set X of elementary possibilities and the binary statement

$$x \perp y,$$

read as “ x and y are mutually exclusive” or “orthogonal.” Such a pair $(X, \perp)$ is an *orthoset*, also called an orthogonality space. An arbitrary orthoset is little more than a simple graph. Vetterlein and collaborators ask what additional conditions make this very sparse structure behave like the set of rays of an inner-product space [4, 5].

The crucial condition is called *linearity*. It is not merely the demand that contexts overlap, nor the demand that every listed context be Boolean. It is a global replacement or Gram–Schmidt-type condition applying to every pair of points. At rank at least four it is strong enough to recover an anisotropic Hermitian space over a division ring with involution. Later conditions concerning automorphisms, transitivity, and local rotations narrow the possible scalar structures [6–8].

The cited orthoset papers state these hypotheses and algebraic conclusions precisely. The purpose of the present comparison is not to correct their theorems or their terminology. It is to place the levels of structure obtained in those theorems alongside partition logics and to distinguish conclusions that arise at different stages of the respective constructions.

The shared vocabulary of contexts and exclusivity can obscure the difference between the two programmes. Partition logic is primarily a concrete representation theory for selected finite tests; the linear-orthoset programme is a

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characterisation theory for complete, highly homogeneous projective orthogonality geometries. Their intersection is important, but neither class contains the other in the naive sense.

This paper develops that statement carefully. Three distinctions will recur:

1. a family of *designated complete tests* is more informative than its pairwise exclusivity graph;
2. a representation by arbitrary orthogonal projections is weaker than a representation by rays, and a finite ray representation is much weaker than being the entire projective space of a Hermitian space;
3. an algebraic Hermitian space over a division ring is not yet the standard real or complex Hilbert space of quantum theory.

II. PARTITION LOGICS

A. Partitions, contexts, and pasted Boolean algebras

Let S be a nonempty finite set. A partition

$$\pi = \{A_1, \dots, A_n\}$$

consists of nonempty, pairwise disjoint cells whose union is S . It generates an equivalence relation

$$s \sim_\pi t \iff s, t \in A_i \text{ for some } i. \quad (1)$$

Conversely, every equivalence relation on S determines a unique partition by its equivalence classes. Operationally, $s \sim_\pi t$ says that the context π supplies no available outcome by which the two underlying cases can be distinguished. Thus the carrier elements may be epistemically inaccessible, or “hidden,” while their equivalence classes are experimentally accessible. The partition generates the Boolean algebra

$$\mathcal{B}(\pi) = \left\{ \bigcup_{i \in I} A_i : I \subseteq \{1, \dots, n\} \right\}. \quad (2)$$

The cells A_i are atoms of this *particular* Boolean algebra. They are the elementary outcomes of the associated measurement context.

No uniformity is required at this level: a family $\mathcal{P} = \{\pi_j : j \in J\}$ may contain partitions with different cardinalities $|\pi_j|$. We call the family *n-uniform* when

$$|\pi_j| = n \quad \text{for every } j \in J. \quad (3)$$

Here uniformity concerns the *number of cells*, not their sizes as subsets of S ; the cardinalities $|A_i|$ may vary. Our quantum-mechanical comparison focuses on this *n-uniform* case because a complete rank-one projective measurement in an *n*-dimensional Hermitian or Hilbert space

contains exactly *n* mutually orthogonal rays. Degenerate projective measurements and general POVMs need not fit this restriction.

For a family of partitions $\mathcal{P} = \{\pi_j : j \in J\}$, define

$$\mathcal{L}(\mathcal{P}) = \bigcup_{j \in J} \mathcal{B}(\pi_j). \quad (4)$$

Boolean operations are certainly available among propositions belonging to a common block $\mathcal{B}(\pi_j)$. Depending on the chosen formalism, $\mathcal{L}(\mathcal{P})$ may be treated as a concrete logic, a partial Boolean algebra, a test space, an orthoalgebra, or an orthomodular poset when the relevant additional axioms hold [3, 9–12].

The same object has several complementary interpretations:

- In a generalized urn, S indexes ball types and a context corresponds to viewing a ball through one colour filter.
- In a finite Mealy automaton, S is the set of possible initial states and a context corresponds to an input experiment; states are grouped according to the output they produce.
- In propositional language, each $\mathcal{B}(\pi_j)$ is a maximal collection of jointly decidable propositions made available by the model.

Generalized urn and finite-automaton partition logics are logically equivalent: their lookup and output functions can be converted into each other explicitly [3].

B. Two-valued states and the classical carrier

Every element $s \in S$ determines a two-valued evaluation on all represented events,

$$v_s(A) = \begin{cases} 1, & s \in A, \\ 0, & s \notin A. \end{cases} \quad (5)$$

Within each Boolean context exactly one cell is assigned value one. Distinct represented events are separated by at least one such valuation. The converse is the standard state-space construction underlying concrete and partition logics [3, 10, 13–15]. For clarity, it can be stated directly at the level of outcomes and tests.

Proposition 1 (Partition-representation criterion). *Let $\mathfrak{T} = (X, \mathcal{E})$ be a finite test hypergraph. It has a faithful set-partition representation by nonempty cells if and only if it possesses a family V of admissible two-valued states such that*

1. for every $x \in X$ there is $v \in V$ with $v(x) = 1$, and
2. for distinct $x, y \in X$ there is $v \in V$ with $v(x) \neq v(y)$.

Here admissibility means that exactly one outcome in each $E \in \mathcal{E}$ has value one.

Proof. Given a partition representation on S , the evaluations v_s from Eq. (5) have both properties because every cell is nonempty and distinct cells are distinct subsets of S . Conversely, take $S = V$ and associate with each outcome x the subset

$$A_x = \{v \in V : v(x) = 1\}. \quad (6)$$

The two conditions make the A_x nonempty and pairwise distinct. Admissibility makes $\{A_x : x \in E\}$ a partition of V for every test $E \in \mathcal{E}$. $\square$

Consequently, a Kochen–Specker-uncolourable test hypergraph cannot be a partition logic. More generally, a finite ray test hypergraph also has a faithful partition representation exactly when its admissible two-valued states contain such a nonzero separating family. In the usual operational reading, this is precisely a separating non-contextual deterministic model for the specified tests. It does not identify the finite ray fragment with a complete linear orthoset. Existence of just one two-valued state is weaker than partition representability.

This property is central. Partition logics may be nondistributive when viewed as pasted empirical structures, but they retain a faithful classical set representation. Their probabilistic states may therefore include the convex hull of two-valued states. The existence of that classical probability model does not prevent the same finite exclusivity graph from also possessing a vector representation with Born-type weights; the empirical graph alone does not determine the probability calculus [16].

C. Cells are local atoms, not necessarily global atoms

A cell of a partition is an atom in its own Boolean block, but it need not be a minimal nonzero element of the entire concrete logic. If A and B belong to different partitions, it can happen that $A \subseteq B$. Hypergraph vertices then represent *outcomes of tests* rather than global lattice atoms.

This observation is important when comparing a concrete partition logic with a Greechie diagram. A Greechie diagram is normally read as a pasting of Boolean blocks whose vertices are global atoms [17]. The same drawing, read merely as a test hypergraph, need not define the same ordered structure. Short loops are especially prone to hiding additional cross-context order relations. Hence “same hypergraph,” “same exclusivity graph,” and “isomorphic empirical logic” are distinct claims.

III. ORTHOSETS AND HERMITIAN PROJECTIVE GEOMETRY

A. Orthogonality and orthogonal closure

Definition 1. An orthoset is a pair $(X, \perp)$, where X is nonempty and $\perp$ is a symmetric, irreflexive relation on X . Its rank is the supremum of the cardinalities of pairwise orthogonal subsets of X .

Thus rank concerns the largest attainable cardinality when the supremum is a maximum. Inclusion-maximal orthogonal subsets need not all have that cardinality unless an additional uniformity condition is imposed.

For $A \subseteq X$, put

$$A^\perp = \{x \in X : x \perp a \text{ for every } a \in A\},$$

$$\text{cl}(A) = A^{\perp\perp}. \quad (7)$$

The orthoclosed subsets

$$\mathcal{C}(X) = \{A : A \subseteq X, A = A^{\perp\perp}\} \quad (8)$$

form a complete ortholattice. If $X = P(H)$ is the set of one-dimensional subspaces of a suitable inner-product space, then $\{e, f\}^{\perp\perp}$ is the projective line spanned by e and f . This motivates defining an abstract line by

$$\ell(e, f) = \{e, f\}^{\perp\perp}, \quad e \neq f. \quad (9)$$

An orthoset by itself does not specify which maximal orthogonal subsets were operationally designated as tests. If contexts are reconstructed as maximal orthogonal subsets or maximal cliques, new “contexts” may appear that were absent from the original test space. Conversely, a maximal orthogonal subset inside a finite induced substructure need not be maximal in an ambient projective geometry.

B. The linearity condition

The decisive postulate in the Vetterlein programme is the following [4, 5].

Definition 2. An orthoset $(X, \perp)$ is linear if, for any distinct $e, f \in X$, there is $g \in X$ such that

$$\{e, f\}^\perp = \{e, g\}^\perp \quad (10)$$

and exactly one of f and g is orthogonal to e .

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The two sides of Eq. (10) have the same double orthogonal closure, so f and g lie on the same abstract

¹ This “exactly one of f and g ” formulation is the linearity condition used by Paseka and Vetterlein [5]; it supplies the replacement step described below.

line through e . If f is not orthogonal to e , the condition supplies an orthogonal replacement g without changing the line. If $f \perp e$, it requires a third, nonorthogonal point on that line. Thus linearity combines a Gram-Schmidt-type replacement property with the demand that abstract lines are sufficiently populated. Merely pasting Boolean contexts along common outcomes does not imply either requirement globally.

C. Representation theorem and the meaning of rank four

A Hermitian space in this literature is a right vector space H over a division ring K equipped with an involution $\alpha \mapsto \alpha^*$ and an anisotropic Hermitian sesquilinear form. Anisotropy means

$$\langle x, x \rangle = 0 \implies x = 0. \quad (11)$$

The projective points $P(H)$ are the one-dimensional subspaces, with orthogonality induced by the form.

The central representation result may be summarized as follows [5]:

Every linear orthoset of rank at least four is, up to orthoisomorphism, the projective orthoset $(P(H), \perp)$ of an anisotropic Hermitian space H ; conversely, such projective orthosets are linear.

The representing Hermitian space is essentially unique in the appropriate semilinear or quasiunitary sense; recent work also studies which maps between orthosets correspond to adjointable or quasilinear maps between Hermitian spaces [18].

Rank four is a geometric threshold, not a physical assertion that three-level quantum systems do not exist. Rank n corresponds to vector-space dimension n and projective dimension $n - 1$. At projective dimension at least three, the classical coordinatization theorem forces abstract projective spaces to arise from vector spaces over division rings. At rank three one has an abstract projective plane, and non-Desarguesian planes show that coordinatizability over a division ring is not automatic. A particular rank-three geometry can of course be Desarguesian; what fails is a general theorem from the stated axioms alone.

Rank two is more degenerate still. In a two-dimensional anisotropic Hermitian space every ray has a unique orthogonal ray. Consequently the bare orthogonality relation looks like a collection of disjoint pairs. It does not, by itself, reveal whether the underlying scalar system was real, complex, or something more general.

D. Hermitian space is not synonymous with Hilbert space

In some elementary texts “Hermitian space” means a finite-dimensional complex inner-product space, which

is automatically a Hilbert space. That is not the broad usage relevant here. Table I records the difference.

For example, $\mathbb{Q}^n$ with the usual dot product is an anisotropic positive quadratic space, but it is not the real Hilbert space $\mathbb{R}^n$: its scalar field and induced metric are incomplete. Completing it adds scalar limits and projective points. Thus generalized Hermitian geometry and the standard complex Hilbert-space formalism are distinct conclusions; passage from the former to the latter requires additional field, topological, and completeness assumptions.

IV. THE SYMMETRY REFINEMENTS IN THE VETTERLEIN PROGRAMME

Linearity yields generalized Hermitian geometry. Further papers investigate what can be forced by symmetry assumptions. The results form a sequence of increasingly specialized characterisations, not a single theorem with “Hilbert space” as an unqualified conclusion.

A. Gradual transitivity

In gradual transitivity, suitable local automorphisms act on the line generated by two points while fixing its orthogonal complement. A divisible subgroup of the circle group supplies roots of all finite orders. Together with linearity, the relevant assumptions lead to positive-definite quadratic spaces over an ordered field; an additional simplicity condition makes the field Archimedean, hence isomorphic to a subfield of $\mathbb{R}$ [6].

Divisibility here is algebraic: each group element admits n th roots. This condition by itself entails neither topological continuity, differentiability, nor metric completeness. Likewise, “subfield of $\mathbb{R}$ ” allows a proper, incomplete subfield.

Related homogeneity and transitivity assumptions likewise lead first to anisotropic Hermitian spaces and, under divisible transitivity, to positive quadratic spaces over ordered fields. Quasiprimitivity can force an embedding of the scalar field into $\mathbb{R}$ [7].

B. Line symmetry, quadratic orthosets, and the Hilbert field

The later approach calls an orthoset line-symmetric when its automorphisms act transitively on the collection of lines and suitable local automorphism groups act transitively within each line. At rank at least four, non-Boolean line-symmetric orthosets correspond to transitive Hermitian spaces. In finite dimension these are standard-form spaces K^n over Pythagorean formally real division rings with involution [8].

A stronger condition requires, for each line, a *commutative* local group acting transitively on that line. Such

TABLE I. Algebraic Hermitian spaces versus standard Hilbert spaces.

Feature	Hermitian space in the representation theorem	Standard Hilbert space
Scalars	A division ring K with involution	$\mathbb{R}$ or $\mathbb{C}$
Form	Anisotropic Hermitian sesquilinear form	Positive-definite real or complex inner product
Topology	Not part of the basic definition	Norm topology induced by the inner product
Completeness	Not required	Every Cauchy sequence converges
Finite-dimensional case	May use a division ring other than $\mathbb{R}$ or $\mathbb{C}$	Over $\mathbb{R}$ or $\mathbb{C}$; metric completeness is automatic
Physical structure	Orthogonality geometry of rays	Still requires a choice of states, observables, composition, and dynamics for quantum theory

orthosets are termed *quadratic*. The resulting spaces are quadratic spaces over commutative ordered fields with the identity involution. Thus this particular route is real-type; it is not a selection principle for complex Hilbert space.

Let $\mathbb{R}_H$ denote the *Hilbert field*, the least Pythagorean subfield of $\mathbb{R}$, equivalently the Pythagorean closure of $\mathbb{Q}$ inside $\mathbb{R}$. It is countable, algebraic over $\mathbb{Q}$, dense in $\mathbb{R}$, and incomplete. For every finite $n \geq 4$,

$$(\mathbb{P}(\mathbb{R}_H^n), \perp) \hookrightarrow (X, \perp) \quad (12)$$

for every non-Boolean quadratic orthoset X of rank n [8]. Equation (12) is a minimality or universal-embedding statement. It does not assert that $X = \mathbb{P}(\mathbb{R}^n)$, that $X = \mathbb{P}(\mathbb{C}^n)$, or that $\mathbb{R}_H$ is complete.

C. Routes toward complex Hilbert space

Earlier finite-rank work employed stronger symmetry and root assumptions and obtained orthomodular spaces over a dense involutive subfield of $\mathbb{C}$, not necessarily over all of $\mathbb{C}$ [19]. For infinite rank, additional lattice and symmetry conditions can invoke the kind of rigidity supplied by Solér's theorem and lead to classical complex Hilbert space [20–22]. Those results are obtained from hypotheses stronger than the bare definition of an orthoset or finite context intertwining. This progression of assumptions is an integral part of the theorem hierarchy.

V. THE COMMON CORE AND THE FORGETFUL PASSAGE

Let X be the collection of cells occurring in a family of partitions. One may define

$$x \perp y \iff x \neq y \text{ and there is } \pi \in \mathcal{P} \text{ such that } x, y \in \pi. \quad (13)$$

Here equal subsets occurring in different partitions are identified as one cell of X ; X is not a set of context–outcome incidence pairs. This identification is what makes a shared outcome a genuine point of intersection between tests. A second possible relation is global set disjointness, $x \perp_S y$ when $x \cap y = \emptyset$, whether or not x and y occur in

a common partition. It is generally larger than Eq. (13). We use the test-induced relation because it records operational co-test exclusivity; the distinction is essential to the additional orthogonal sets considered below.

This constructs an orthoset from the test hypergraph. It is a useful bridge, but it is a *forgetful* construction:

$$\begin{aligned} \text{partition family} &\longrightarrow \text{test hypergraph} \\ &\xrightarrow{\text{forget tests}} (X, \perp) \longrightarrow \mathcal{C}(X). \end{aligned} \quad (14)$$

The first structure remembers the carrier set and its set inclusions. The test hypergraph remembers which collections are complete tests. The orthoset remembers only pairwise exclusivity. The closure lattice $\mathcal{C}(X)$ is then reconstructed from the pairwise relation and need not coincide with the original union of pasted Boolean algebras.

The similarities are nevertheless substantive. Both approaches isolate Boolean contexts, represent mutual exclusivity by orthogonality, use rank as the maximal number of mutually exclusive outcomes, and study how contexts intertwine. Finite partition logics may occur as finite subconfigurations of projective orthogonality geometries, and one and the same exclusivity graph may admit both a set-theoretic and a faithful vector representation [16]. The shared exclusivity graph alone neither makes every partition logic a projective geometry nor determines a projective completion from an arbitrary finite family of intertwined contexts.

VI. RESTRICTING AND ENLARGING PARTITION LOGICS TOWARD ORTHOSETS

There is a natural route from a partition logic toward an orthoset, but there are several inequivalent operations that may be called “completion.” Starting with a finite carrier S and a partition family $\mathcal{P}$, one may successively forget structure, add tests, add events, or add points. These operations have different mathematical and operational effects and will therefore be treated separately.

TABLE II. Comparison of the information carried by the two approaches.

Question	Partition logic	Linear-orthoset programme
Primitive data	Finite carrier S and selected partitions of S	Points X and a symmetric, irreflexive relation $\perp$
Operational tests	Explicitly designated hyperedges/partitions	Usually inferred from orthogonality and closure
Events	Unions of partition cells, with partial Boolean operations	Orthoclosed subsets $A = A^{\perp\perp}$ are constructed secondarily
Classical valuations	Separating evaluations inherited from $s \in S$	Not guaranteed; projective Hilbert geometries in dimension ≥ 3 generally obstruct global two-valued valuations
Homogeneity	None required; contexts may be irregular	Strong global linearity and, in later results, transitivity or local rotation groups
Typical size	Finite in the standard automaton/urn setting	A linear orthoset of rank at least three is infinite
Representation goal	Realize chosen finite tests by set partitions	Characterize the complete ray orthogonality geometry of a Hermitian space
Probabilities	Classical convex hull is naturally available	Orthogonality alone does not choose a probability rule
Immediate result	Finite contextual set representation	Generalized Hermitian geometry; standard complex Hilbert structure requires further hypotheses

A. The intermediate structure: exclusivity plus designated tests

Let X_0 be the set of distinct cells occurring in $\mathcal{P}$, let $\perp$ be the relation in Eq. (13), and let

$$\mathcal{E}_0 = \{E_\pi : E_\pi \text{ is the set of cells of } \pi, \pi \in \mathcal{P}\}. \quad (15)$$

The triple

$$\mathfrak{T}_0 = (X_0, \perp, \mathcal{E}_0) \quad (16)$$

retains both pairwise exclusivity and the family of operationally designated complete tests. It is an orthogonality test space in the sense that every $E \in \mathcal{E}_0$ is a set of pairwise orthogonal outcomes; test-space formalisms take precisely such outcome-test incidence as primitive [9, 11, 12].

A probabilistic state on this structure is a map $p : X_0 \rightarrow [0, 1]$ satisfying

$$\sum_{x \in E} p(x) = 1 \quad (E \in \mathcal{E}_0). \quad (17)$$

For a total two-valued state, $p(x) \in \{0, 1\}$, Eq. (17) is the usual admissibility demand that exactly one outcome in every complete context be true. Thus exclusivity is carried by $\perp$, whereas operational completeness is carried by $\mathcal{E}_0$ and admissibility constrains states relative to those tests. Completeness here means exhaustiveness of a test; it is not metric completeness of a Hilbert space and not the closure condition $A = A^{\perp\perp}$.

Forgetting $\mathcal{E}_0$ leaves the bare orthoset $(X_0, \perp)$. This restriction is exactly the minimalistic step taken in the linear-orthoset programme: mutual exclusivity remains, but the original list of apparatus settings does not [4, 5].

B. Test saturation: more contexts, fewer states

Assume that the original partitions are n -uniform and that the intended rank is n . One possible enlargement declares every orthogonal n -tuple to be a complete test:

$$\mathcal{E}_{\text{sat}}^{(n)} = \{E \subseteq X_0 : |E| = n \text{ and } x \perp y \text{ for all distinct } x, y \in E\}. \quad (18)$$

At rank n these are maximum orthogonal sets. Without a fixed uniform rank, one might instead use all inclusion-maximal orthogonal sets. Either convention is an additional *completeness postulate*; maximality in an abstract graph does not by itself say that an apparatus implements the corresponding test.

This construction exhibits a useful reversal. It enlarges the family of tests, $\mathcal{E}_0 \subseteq \mathcal{E}_{\text{sat}}^{(n)}$, but restricts the state space:

$$\Omega(\mathcal{E}) = \left\{ p \in [0, 1]^{X_0} : \sum_{x \in E} p(x) = 1 \text{ for every } E \in \mathcal{E} \right\}, \quad (19)$$

$$\Omega(\mathcal{E}_{\text{sat}}^{(n)}) \subseteq \Omega(\mathcal{E}_0).$$

For the triangle of Sec. VIII, pairwise exclusivity makes

$$E_\Delta = \{a, c, e\} \quad (20)$$

an orthogonal triple even though it is not a designated partition. Saturation adds E_Δ and hence the equation $p(a) + p(c) + p(e) = 1$. The classical valuation v_4 associated with the underlying case $4 \in S$ assigns zero to all three singleton cells $a = \{1\}$, $c = \{3\}$, and $e = \{2\}$.

Consequently v_4 is admissible for the original three partitions but not for the saturated test family. The “phantom context” therefore has an empirically nontrivial effect: adding it removes a classical state.

This example also shows that exclusivity plus admissibility does not determine the original partition model. Admissibility acquires determinate content only after the complete orthogonal sets have been specified.

C. Orthogonal closure: enlarging the event language

A different construction keeps the point set fixed and forms

$$\mathcal{C}(X_0) = \{A \subseteq X_0 : A = A^{\perp\perp}\}. \quad (21)$$

For any orthoset this is a complete ortholattice. For a linear orthoset its singletons are the atoms and the lattice has the additional projective properties used in the Hermitian representation theorem [5]. This is canonical once $\perp$ has been chosen, but it adds *events*, not points. Moreover, $\mathcal{C}(X_0)$ is a new event structure over the cells X_0 ; it need not be identical to, or even a literal superset of, the concrete subset logic $\mathcal{L}(\mathcal{P})$ over the carrier S .

Test saturation and orthogonal closure are also independent. The former adds normalization equations by designating further tests. The latter constructs fixed points of a Galois closure from pairwise orthogonality. Neither operation alone makes an arbitrary finite orthoset linear or populates its abstract projective lines with the missing points.

D. Point enlargement and projective completion

The strongest construction seeks a larger orthoset $(\hat{X}, \perp_{\hat{X}})$ and an injection $i : X_0 \rightarrow \hat{X}$ such that

$$x \perp y \iff i(x) \perp_{\hat{X}} i(y). \quad (22)$$

One then asks that $\hat{X}$ have the intended rank and satisfy linearity, Eq. (10). Linearity is the decisive replacement principle: it requires enough points on every abstract line to reproduce the orthogonal-projection step. At rank at least four it makes $\hat{X}$ isomorphic to the complete ray orthoset $P(H)$ of an anisotropic Hermitian space. Further symmetry, field, and topological assumptions remain separate.

Such a point completion is not supplied canonically by an arbitrary finite partition logic. It may fail to exist with the requested rank and with every original context remaining a complete rank-one test. The triangle proves this failure at rank three. If the dimension is raised to four, its exclusivity graph has the faithful ray representation of Eq. (35), but the displayed triples cease to be complete bases. Equations (36) and (37) give one explicit enlargement by three new rays. Different ambient fields,

forms, dimensions, and embeddings can moreover lead to inequivalent completions. A definite projective completion therefore requires a specified category of embeddings or a genuine universal property.

E. A disciplined hierarchy

A conservative way to generalize partition logics is consequently to retain the layers

$$\begin{aligned} (S, \mathcal{P}) &\longrightarrow (X_0, \perp, \mathcal{E}_0) \longrightarrow (X_0, \perp) \\ &\longrightarrow \mathcal{C}(X_0) \text{ or } (\hat{X}, \perp_{\hat{X}}). \end{aligned} \quad (23)$$

The first arrow forgets underlying cases and set inclusions but preserves designated tests. The second forgets which tests were operationally given. Test saturation may enlarge $\mathcal{E}_0$ and thereby restrict admissible states; orthogonal closure enlarges the event language; a projective completion instead adjoins points and imposes linearity. In this precise test-space sense, *exclusivity plus designated completeness plus admissibility* is a useful bridge from partition logic toward quantum-like models. Vetterlein’s full reconstruction additionally imposes global linearity, followed, where desired, by symmetry and completeness assumptions of other kinds.

VII. FOUR DIFFERENT NOTIONS OF REPRESENTATION

Much apparent disagreement disappears once the word “representation” is qualified. At least four notions are relevant.

A. Set-partition representation

Cells are nonempty subsets of a finite S , contexts are partitions of S , and events are unions of cells. Orthogonality is disjointness within a test, or globally if the concrete set representation is used. This is the defining representation for a partition logic.

B. Projection representation

Every finite set-partition model has an immediate Hilbert-lattice embedding. This is the familiar coordinate representation of a finite Boolean algebra, included here only to distinguish projection representability from the stronger ray and projective notions below.

Remark 1 (Diagonal projection representation). *Let S be finite and let $\mathcal{P}$ be any family of partitions of S . Put $H = \ell^2(S)$ with orthonormal basis $\{e_s : s \in S\}$. For every represented event $A \subseteq S$, define*

$$H_A = \text{span}\{e_s : s \in A\}, \quad P_A = \sum_{s \in A} |e_s\rangle\langle e_s|. \quad (24)$$

Then $A \mapsto H_A$, equivalently $A \mapsto P_A$, is injective, preserves orthocomplementation and inclusion, and maps every partition $\{A_1, \dots, A_n\}$ to an orthogonal direct-sum decomposition

$$H = H_{A_1} \oplus \dots \oplus H_{A_n}, \quad I = P_{A_1} + \dots + P_{A_n}. \quad (25)$$

Disjoint subsets span orthogonal coordinate subspaces, unions correspond to orthogonal sums, and complements correspond to orthogonal complements. Since the coordinate basis vectors are distinct, the map is injective.

All projections in this construction commute. It therefore proves Hilbert *representability* in a weak algebraic sense, not intrinsic quantum noncommutativity. Moreover, a cell containing several elements of S becomes a higher-rank projector.

C. Faithful ray representation of an exclusivity graph

Given an exclusivity graph $G = (V, E)$, a faithful orthogonal representation in K^d assigns a distinct ray r_v to every vertex such that

$$\{u, v\} \in E \iff r_u \perp r_v, \quad (26)$$

up to the convention of replacing G by its complement. Such a representation may exist in a dimension larger than the number of outcomes in each displayed context. The images of a context are then mutually orthogonal but need not constitute a complete basis. Faithfulness of the graph also does not by itself recover which maximal cliques were operationally designated as tests. This is related to the orthonormal-representation machinery of Lovász and its use in graph-theoretic contextuality, but Eq. (26) additionally requires nonedges to remain nonorthogonal and therefore specifies a faithful induced representation [12, 16, 23, 24].

D. Full Hermitian projective representation

The strongest assertion is an isomorphism

$$(X, \perp) \cong (P(H), \perp), \quad (27)$$

where X contains *all* projective points of H and the abstract closure agrees with projective span. This is the conclusion of the rank- ≥ 4 linear-orthoset representation theorem. A finite ray configuration inside $P(H)$ is not Eq. (27); it is only a fragment, and its internal double-orthogonal closure may differ from closure in the ambient projective space.

VIII. THE THREE-UNIFORM TRIANGLE

The triangle is an economical example separating all four notions. Let $S = \{1, 2, 3, 4\}$ and consider the three

partitions

$$\begin{aligned} \pi_1 &= \{\{1\}, \{2, 4\}, \{3\}\}, \\ \pi_2 &= \{\{3\}, \{1, 4\}, \{2\}\}, \\ \pi_3 &= \{\{2\}, \{3, 4\}, \{1\}\}. \end{aligned} \quad (28)$$

Label their six distinct cells by

$$\begin{aligned} a &= \{1\}, & b &= \{2, 4\}, & c &= \{3\}, \\ d &= \{1, 4\}, & e &= \{2\}, & f &= \{3, 4\}. \end{aligned} \quad (29)$$

The contexts are

$$\{a, b, c\}, \quad \{c, d, e\}, \quad \{e, f, a\}. \quad (30)$$

A. A valid partition model with extra cross-context order

Equation (28) is already a concrete partition representation. It also reveals relations not visible in Fig. 1:

$$\begin{aligned} e &= \{2\} \subset b = \{2, 4\}, \\ a &= \{1\} \subset d = \{1, 4\}, \\ c &= \{3\} \subset f = \{3, 4\}. \end{aligned} \quad (31)$$

Thus all six vertices are atoms of their local three-element partitions, but only a, c, e are singleton atoms in the global Boolean algebra 2^S . Reading the drawing as an abstract Greechie diagram of six global atoms would therefore produce a different ordered object from the concrete subset logic.

Furthermore, a, c, e occur pairwise in the three different contexts, so the orthoset obtained from Eq. (13) regards them as a pairwise orthogonal triple. Within the six-vertex orthoset it is maximal, even though it was not stipulated as a complete test. Pairwise exclusivity has therefore generated a fourth “phantom context.” In the underlying set model, $a \cup c \cup e = \{1, 2, 3\}$ leaves the case 4 unaccounted for; the triple is jointly disjoint but is not a partition of S .

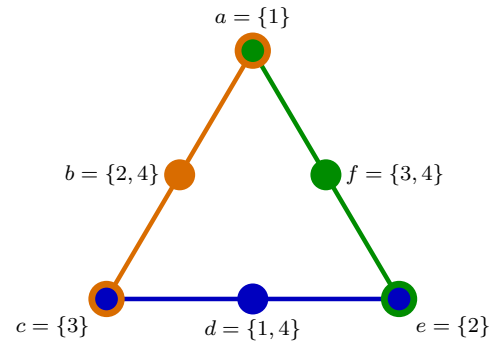


FIG. 1. The three-uniform triangle test hypergraph. Each colored side is one designated three-outcome context; concentric colors mark outcomes shared by two contexts. The corner vertices a, c, e are pairwise exclusive, although $\{a, c, e\}$ is not one of the designated partitions.

B. A general rank obstruction and the triangle

The incidence pattern in Fig. 1 is a loop of order three. Such a loop is excluded from a Greechie diagram representing an orthomodular poset; loops of order four mark the corresponding obstruction to the lattice property. These short-loop phenomena and their Hilbert-lattice realizability have long been part of the pasting literature [15, 17]. The following elementary proposition does not claim a new loop lemma. It isolates the dimension-theoretic mechanism in orthoset language and states the resulting obstruction for arbitrary rank.

Proposition 2 (Unique-complement obstruction). *Suppose that an exclusivity structure $(X, \perp)$ contains a pairwise orthogonal set $B = \{x_1, \dots, x_n\}$. Suppose also that for some j there is a point $y \notin B$ satisfying*

$$y \perp x_i \quad \text{for every } i \neq j. \quad (32)$$

Then no assignment of distinct rays to the points of X in an n -dimensional anisotropic Hermitian space can preserve all the stated orthogonalities.

Proof. The images of $x_1, \dots, x_n$ are n mutually orthogonal rays and hence form an orthogonal basis spanning the entire n -dimensional space. The orthogonal complement of the span of $\{x_i : i \neq j\}$ is therefore the one-dimensional subspace represented by the ray x_j . Condition (32) forces the nonzero image of y into that same one-dimensional subspace, so the image rays of y and x_j coincide. Because $y \notin B$, this contradicts faithfulness. $\square$

Equivalently, a necessary combinatorial condition for faithful rank- n realizability is that no vertex outside an orthogonal n -clique have all $n - 1$ vertices of one of its codimension-one faces as neighbours. The condition is only necessary; passing it does not guarantee a global vector realization.

Corollary 1 (Triangle obstruction). *There is no assignment of six distinct projective points of a three-dimensional anisotropic Hermitian space to the vertices in Eq. (30) that preserves the pairwise orthogonalities within the displayed contexts.*

Proof. The displayed contexts imply

$$a \perp c, \quad c \perp e, \quad e \perp a, \quad (33)$$

so $B = \{a, c, e\}$ is an orthogonal triple. The vertex $b \notin B$ is orthogonal to a and c , and Proposition 2 forces $b = e$. Cyclically, the same argument gives

$$b = e, \quad d = a, \quad f = c. \quad (34)$$

Thus the representation is not faithful. $\square$

Only anisotropy and dimension enter this argument, not a specifically real or complex scalar field. The class of finite partition test hypergraphs is therefore not contained in the class of finite ray test hypergraphs faithfully realizable at the same rank.

C. What representations remain possible?

First, the exclusivity graph does have a faithful ray representation already in $\mathbb{R}^4$, and hence also after complexification. With $e_1, \dots, e_4$ the standard basis, take

$$\begin{aligned} a &= e_1, & b &= \frac{1}{\sqrt{2}}(e_2 + e_4), & c &= e_3, \\ d &= \frac{1}{\sqrt{2}}(e_1 + e_4), & e &= e_2, & f &= \frac{1}{\sqrt{2}}(e_3 + e_4). \end{aligned} \quad (35)$$

Each displayed triple is orthonormal, and among these six rays orthogonality occurs exactly for pairs joined within a displayed context. But the triples are not complete bases of $\mathbb{R}^4$; each lacks a fourth ray. Hence this is a faithful representation of the graph as a finite fragment, not a realization of the three hyperedges as complete rank-one contexts.

The missing rays can be adjoined explicitly:

$$\begin{aligned} g &= \frac{1}{\sqrt{2}}(e_2 - e_4), & h &= \frac{1}{\sqrt{2}}(e_1 - e_4), \\ k &= \frac{1}{\sqrt{2}}(e_3 - e_4). \end{aligned} \quad (36)$$

Then

$$\{a, b, c, g\}, \quad \{c, d, e, h\}, \quad \{e, f, a, k\} \quad (37)$$

are orthonormal bases of $\mathbb{R}^4$. This is a genuine enlargement of the test hypergraph: each original three-outcome test becomes a proper subset of a new four-outcome test. It does not turn the original triples into complete three-outcome contexts.

Second, Eq. (24) gives a complete projection-valued representation in $\mathbb{C}^4$:

$$\begin{aligned} \pi_1 &: P_{\{1\}} + P_{\{2,4\}} + P_{\{3\}} = I, \\ \pi_2 &: P_{\{3\}} + P_{\{1,4\}} + P_{\{2\}} = I, \\ \pi_3 &: P_{\{2\}} + P_{\{3,4\}} + P_{\{1\}} = I. \end{aligned} \quad (38)$$

One projector in each row has rank two. All six projectors commute because they are diagonal in the same basis. Together with the ray construction, this gives the following qualified statement:

The triangle occurs as a finite commuting projection configuration, and its exclusivity graph occurs as an incomplete ray configuration in dimension four that can be enlarged to four-outcome bases; it does not occur faithfully as three rank-three ray contexts.

IX. FINITENESS: OBSTRUCTION, BOUNDARY, AND COMPLETION

Two uses of the word “finite” must be separated:

1. *finite rank*: at most n mutually orthogonal points exist;

2. *finite cardinality*: the total point set X is finite.

A finite-dimensional real or complex Hilbert space has finite rank but infinitely many rays. Thus finite rank never implies a finite modality or point set.

The finite analysis of Emir, Kruml, Paseka, and Vetterlein, and its later extensions, yields a sharp fact: a linear orthoset of rank at least three is infinite [4, 25]. Finite rank-two linear orthosets reduce to orthogonal pairings, while finite rank-three structures can satisfy weaker prelinearity properties but not full linearity.

Therefore a finite partition logic of rank at least three cannot itself be the entire linear orthoset $P(H)$. This is consistent with the scope of the representation theorem: a finite contextual structure can at most be a fragment or a structure to be completed. The completion must add enough points to populate projective lines and satisfy the global replacement axiom.

Three further consequences clarify this boundary.

First, completion is substantive, not merely a change of notation. If $X \subset P(H)$ is finite, then the internal closure

$$\{e, f\}^{\perp_X \perp_X} \quad (39)$$

can be much smaller than the full projective line $P(\text{span}(e, f))$. Adding the missing points changes the orthoclosed-event structure.

Second, orthogonality data need not determine a unique completion unless the projective and symmetry axioms required by the representation theorem are assumed. Even after a generalized Hermitian space is obtained, the scalar division ring need not be $\mathbb{C}$.

Third, finiteness alone does not distinguish classical from quantum representability. Every finite partition model embeds diagonally into a finite complex Hilbert space, while only some exclusivity graphs admit faithful ray representations in a prescribed low dimension. What finiteness obstructs is the identification of the finite structure with a *complete linear projective orthoset* of rank at least three.

X. STATES AND PROBABILITIES

Neither a test hypergraph nor an orthogonality graph uniquely fixes its state space. A partition representation supplies a natural classical choice. If $\lambda_s \geq 0$ and $\sum_s \lambda_s = 1$, then

$$p(A) = \sum_{s \in A} \lambda_s \quad (40)$$

is additive on every partition context. These probabilities form the convex hull of the two-valued evaluations in Eq. (5).

A ray representation supplies a different possible resource. Given a unit state vector ψ and ray projectors P_x , one may define

$$p_\psi(x) = \langle \psi, P_x \psi \rangle. \quad (41)$$

For a complete orthonormal basis the weights sum to one. Gleason's theorem shows, in Hilbert-space dimension at least three and under its additivity hypotheses on the full Hilbert lattice, how density operators characterize such measures [26]. But Eq. (41) uses structure not contained in an arbitrary finite graph: a chosen scalar field, inner product, complete contexts, and a state.

This is why one finite exclusivity graph may support both a classical partition model and a quantum vector model with different convex sets of probabilities [16]. Orthogonality is a constraint on joint zero transition probabilities; it is not by itself the Born rule.

The basic linear-orthoset theorem likewise does not reconstruct mixed states, tensor products, composition of systems, time evolution, observable algebras, or operational continuity. Those may be introduced or characterized in richer frameworks; they are additional structures rather than consequences of the bare orthogonality relation.

XI. STRUCTURAL IMPLICATIONS AND SCOPE

The comparison supports the following affirmative statements.

1. Partition logics and orthosets share a combinatorial core of elementary outcomes and exclusivity.
2. A partition test space produces an orthoset by forgetting all but pairwise exclusivity witnessed within tests.
3. The layered structure $(X, \perp, \mathcal{E})$ can preserve both exclusivity and designated complete tests; enlarging $\mathcal{E}$ systematically restricts the admissible state space.
4. A finite test hypergraph is faithfully partition-representable exactly when it has a nonzero separating family of admissible two-valued states.
5. Some partition exclusivity graphs possess faithful orthogonal vector representations; the same graph can consequently admit distinct physical and probabilistic realizations.
6. Vetterlein's linearity axiom identifies when orthogonality is rich enough to recover projective span and generalized Hermitian geometry at rank at least four.
7. Additional automorphism axioms can substantially restrict the allowed Hermitian or quadratic spaces.

The corresponding scope distinctions are as follows.

1. Overlapping or "intertwining" contexts alone do not imply linearity.
2. The exclusivity graph does not retain the complete-test hypergraph in general.

3. Declaring every maximal orthogonal set complete is an additional postulate and can eliminate states inherited from the partition carrier. Orthogonal closure, test saturation, and projective point completion are distinct operations.
4. The unique-complement obstruction in Proposition 2 is a necessary condition for faithful rank- n ray realizability; the three-loop triangle violates it.
5. A Kochen–Specker-uncolourable test hypergraph cannot be a partition logic, because it lacks even one admissible two-valued state.
6. A finite partition logic of rank at least three cannot equal a full linear orthoset, because such a linear orthoset must be infinite.
7. A generalized anisotropic Hermitian space is not automatically a real or complex Hilbert space.
8. The Hilbert-field minimality theorem gives a countable dense real subgeometry that embeds in every quadratic orthoset of the same finite rank; it does not give equality with real or complex projective Hilbert space.
9. Orthogonality and its automorphisms alone do not automatically provide the complete probabilistic and dynamical formalism of quantum mechanics.

In this manuscript, “reconstruction” refers to a representation or characterisation theorem: a relational structure satisfying specified strong axioms is shown to be isomorphic to the projective orthogonality structure of a member of a stated class of inner-product spaces. The operational motivation of the axioms, selection of a scalar field, and construction of the remaining quantum formalism are further questions beyond that representation statement.

XII. CONCLUSION

Partition logics and linear orthosets begin near the same operational image—a collection of experimental alternatives grouped into mutually exclusive contexts—but travel in opposite mathematical directions. Partition logic adds a finite classical carrier beneath the contexts. Vetterlein’s programme removes almost everything except orthogonality and then imposes sufficiently strong global regularity to recover projective geometry.

An intermediate test-orthoset layer makes a controlled interpolation possible. It preserves a set of outcomes, their pairwise exclusivity, and a separately designated family of complete tests. Saturating that family adds normalization conditions and restricts admissible states; closing under $\perp\perp$ builds a new event lattice; adjoining points to enforce linearity seeks a genuinely projective completion. These three enlargements answer different questions.

The relation is therefore neither identity nor disjointness. A finite partition logic can supply the same exclusivity skeleton as a finite fragment of a Hermitian projective geometry, and it always has a diagonal representation by commuting Hilbert-space projections. Yet partition logics can also contain test structures, such as the three-uniform triangle, that cannot be interpreted faithfully by distinct rays at the corresponding rank. The general unique-complement obstruction identifies one combinatorial source of such failures. Conversely, the separating-state criterion identifies exactly when a finite test hypergraph has a faithful partition representation. When only pairwise orthogonality is retained, even the stipulated set of tests may change.

Conversely, a full projective orthoset is vastly richer than a finite pasted family of tests. Its lines are saturated, its orthogonal closure determines projective span, and at rank at least three its linearity forces infinitely many points. Rank four enters because it is the general coordinatization threshold. Symmetry conditions may then restrict the scalar system, with the outcome depending on the hypotheses: an anisotropic Hermitian space, a real-type quadratic space over an ordered or Pythagorean field, a Hilbert-field subgeometry, or—under stronger infinite-dimensional assumptions—a classical Hilbert space.

Thus finite partition logics are useful both as models and as diagnostic comparison cases. They show precisely which consequences follow from local Booleanity and intertwining, and which require the global projective, homogeneity, field, and completeness assumptions of a Hilbert-space reconstruction.

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